

# Docking Universal - Docking Study Report

Report generator version 0.7.0

Target: 1HVR

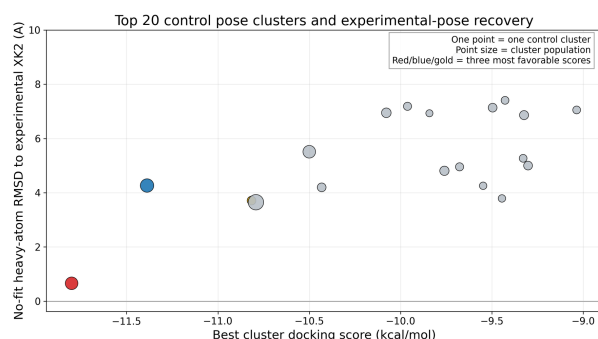
## 1. Configured docking protocol

This table identifies the reusable target-matched protocol, records whether it was approved during this run or reused from an earlier run, and lists its locked docking settings.

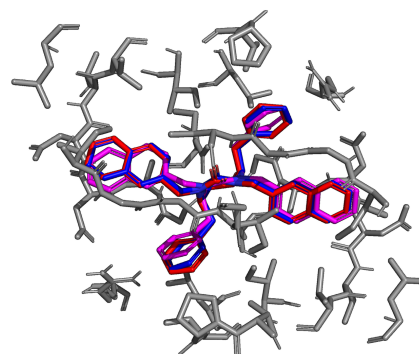
| Approved docking protocol   | Selected value                                        |
|-----------------------------|-------------------------------------------------------|
| Protocol file name          | 1HVR_XK2_control-validated_20260911_140028.duprotocol |
| Approval provenance         | Approved previously and reused for this run           |
| Control approval date (UTC) | 2026-09-11T18:19:44.928063+00:00                      |
| Prepared receptor           | 1HVR.pdbqt                                            |
| Locked docking box / pocket | 1HVR_pocket1.conf                                     |
| Engine                      | vina                                                  |
| Tier                        | repeatability                                         |
| Exhaustiveness              | 16                                                    |
| Modes per job               | 15                                                    |
| Conformers per state        | 3                                                     |
| Independent seeds           | 5                                                     |
| Charge model                | gasteiger                                             |
| pH                          | 7.4                                                   |
| Conformer force field       | mmff94                                                |
| Tautomers enumerated        | True                                                  |
| Conformer RMSD pruning      | 0.75 A                                                |

## Control pose-recovery results

A



B



**Figure 1. Retrospective control performance and pose recovery.** (A) Control-cluster docking score versus symmetry-aware, no-fit heavy-atom RMSD to experimental XK2 in the receptor coordinate frame. (B) Experimental ligand (magenta), lowest-energy pose (red), and lowest-RMSD pose (blue) superimposed in that frame; receptor residues within 5 Å of the displayed ligands are gray.

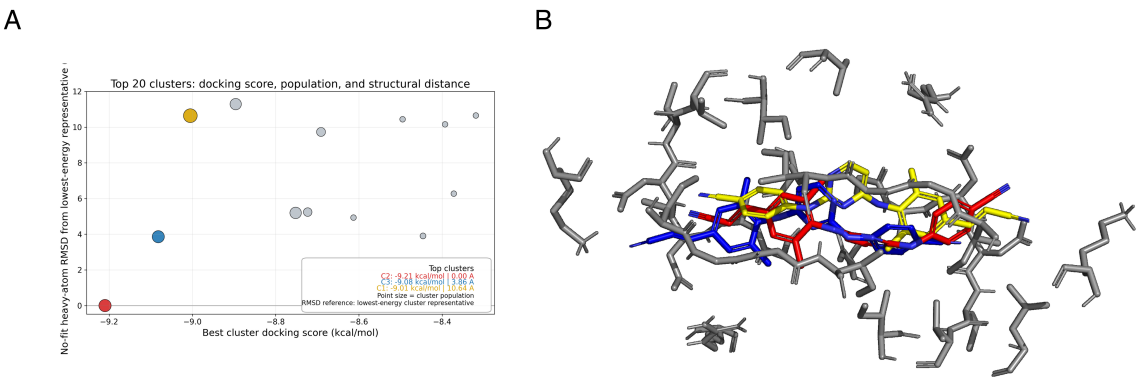
## 2. Ligand docking results

### Summary of docking results

| Summary of docking results  | Recorded result |
|-----------------------------|-----------------|
| Ligand docked               | Rilpivirine     |
| Completion status           | COMPLETED       |
| Docking sites evaluated     | 1               |
| Best retained docking score | -9.21 kcal/mol  |
| Top-ranked cluster          | 2               |
| Top-cluster population      | 38              |
| Independent-seed support    | 5               |
| Conformer support           | 3               |
| Distinct retained clusters  | 13              |
| Selected representatives    | 3               |

### Target: 1HVR | Ligand: Rilpivirine

Docking used the existing approved target-matched protocol shown above. More favorable docking scores are more negative; RMSD and cluster population are separate measures.

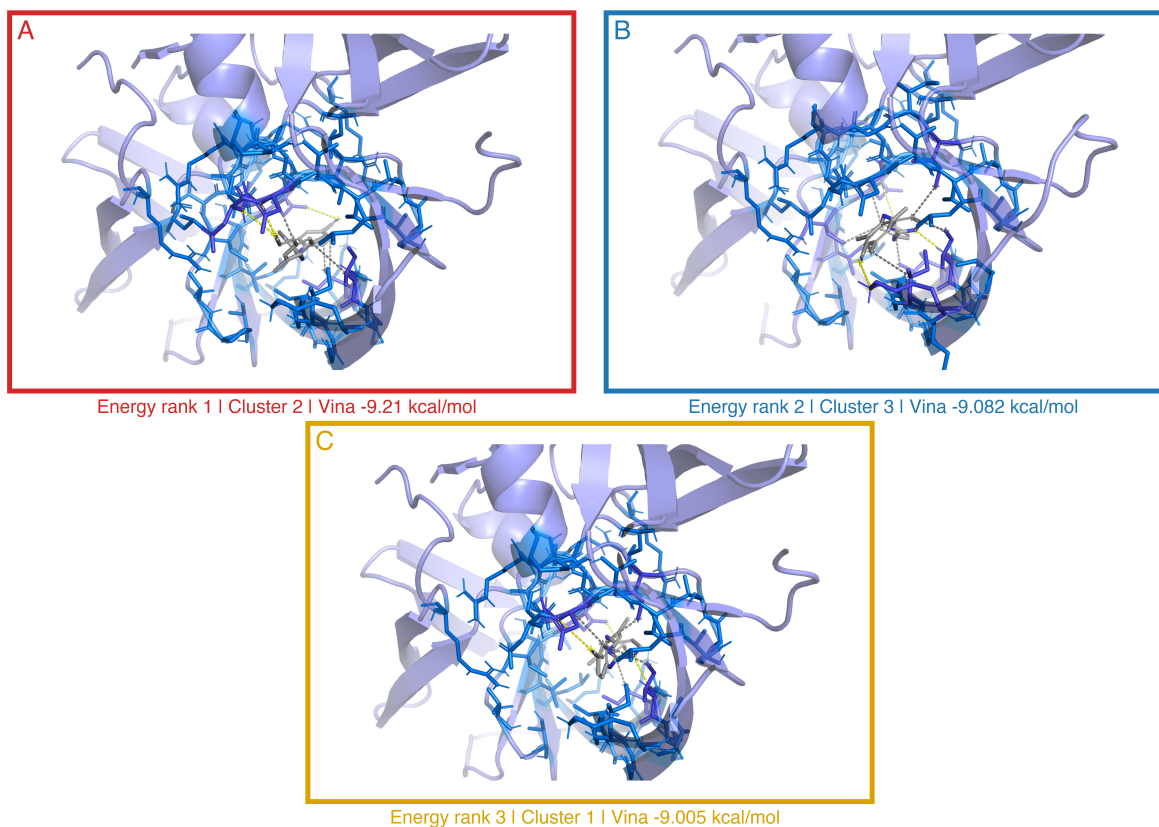


**Figure 2. Docking pose-cluster analysis for Rilpivirine.** (A) Docking score versus symmetry-aware, no-fit heavy-atom RMSD from the lowest-energy cluster representative in the receptor coordinate frame; point size denotes cluster population. (B) Representative structures from the highlighted clusters, using matching cluster colors.

## Clusters represented in the docking plot

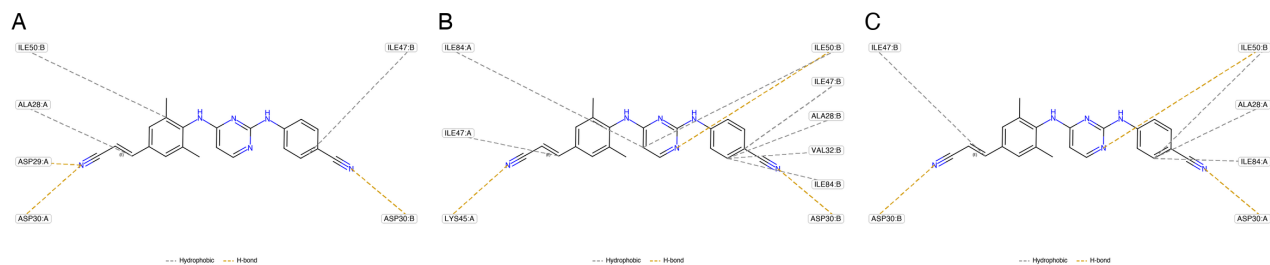
| Rank | Cluster    | Docking score | RMSD (Å)  | Population |
|------|------------|---------------|-----------|------------|
| 1    | Cluster 2  | -9.21         | 0.000000  | 38         |
| 2    | Cluster 3  | -9.082        | 3.858486  | 37         |
| 3    | Cluster 1  | -9.005        | 10.640349 | 50         |
| 4    | Cluster 4  | -8.896        | 11.288869 | 31         |
| 5    | Cluster 5  | -8.752        | 5.192835  | 32         |
| 6    | Cluster 8  | -8.723        | 5.238141  | 14         |
| 7    | Cluster 7  | -8.691        | 9.729086  | 16         |
| 8    | Cluster 6  | -8.613        | 4.929183  | 1          |
| 9    | Cluster 10 | -8.495        | 10.439236 | 1          |
| 10   | Cluster 12 | -8.446        | 3.902452  | 2          |
| 11   | Cluster 13 | -8.393        | 10.154944 | 1          |
| 12   | Cluster 9  | -8.372        | 6.272070  | 1          |
| 13   | Cluster 11 | -8.319        | 10.650684 | 1          |

## Energy-ranked 3D cluster representatives



**Figure 3. Three-dimensional interaction snapshots for Rilpivirine.** Shown are 3 energy-ranked distinct cluster representatives, ordered by docking score. Panels A, B, and C show energy ranks 1, 2, and 3, respectively. Red, blue, and gold match the highlighted clusters in Figure 2. These views support structural inspection; docking score rank does not establish pose correctness.

## Selected 2D pose interaction diagrams



**Figure 4. SDF-aware PLIP interaction diagrams for Rilpivirine.** A, B, and C are the red, blue, and gold energy-ranked cluster representatives shown above. Ligand chemistry comes from each retained SDF; interaction calls come from the retained PLIP report.xml.

### Interpretation and limitations

Docking scores are ranking estimates, not measured binding free energies. Rigid-receptor docking does not model induced fit. Cluster population and seed support describe computational convergence, not biological correctness. Protonation, tautomer, receptor preparation, and box choices can affect results. Experimental validation remains necessary.

### 3. Reproducibility, software, and references

Receptor preparation: strict Meeko; PDBFixer was not needed. AutoDock Vina produced docking scores and poses. PLIP supplied rule-based protein-ligand interaction calls, with retained PLIP XML as the authoritative interaction record. RDKit supplied symmetry-aware heavy-atom RMSD calculations and Butina clustering at a 2.0 Å cutoff. PyMOL produced the molecular panels, and ReportLab assembled this PDF.

**CCD/MODRES audit:** A:67 CSO → CYS (resolved by an available Meeko template); B:67 CSO → CYS (resolved by an available Meeko template).

#### Control-to-new-run software check

**Checked: no software version differences detected.**

| Software          | Control              | New run              | Result |
|-------------------|----------------------|----------------------|--------|
| Docking Universal | 0.7.0                | 0.7.0                | SAME   |
| Python            | 3.9.23               | 3.9.23               | SAME   |
| RDKit             | 2023.9.6             | 2023.9.6             | SAME   |
| MolScrub          | 0.2.2                | 0.2.2                | SAME   |
| Meeko             | 0.7.1                | 0.7.1                | SAME   |
| AutoDock Vina     | AutoDock Vina v1.2.7 | AutoDock Vina v1.2.7 | SAME   |

#### Additional software used for analysis and reporting

| Result element                    | Software   | Version used |
|-----------------------------------|------------|--------------|
| Molecular conversion/PLIP backend | Open Babel | 3.1.0        |
| Interaction calls                 | PLIP       | 2.3.1        |
| 3D rendering                      | PyMOL      | 3.0.0        |
| Plots                             | Matplotlib | 3.9.4        |
| PDF generation                    | ReportLab  | 5.0.0        |

#### Scientific and software references

1. Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. J Chem Inf Model. 2021;61:3891-3898. <https://doi.org/10.1021/acs.jcim.1c00203>
2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
3. Salentin S, Schreiber S, Haupt VJ, Adasme MF, Schroeder M. PLIP: fully automated protein-ligand interaction profiler. Nucleic Acids Res. 2015;43:W443-W447. <https://doi.org/10.1093/nar/gkv315>
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6. RDKit: Open-source cheminformatics software. <https://www.rdkit.org/>
7. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>
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9. ReportLab: PDF generation toolkit and user guide. <https://docs.reportlab.com/>
10. Python Software Foundation. Python Language Reference. <https://www.python.org/>